Week 1

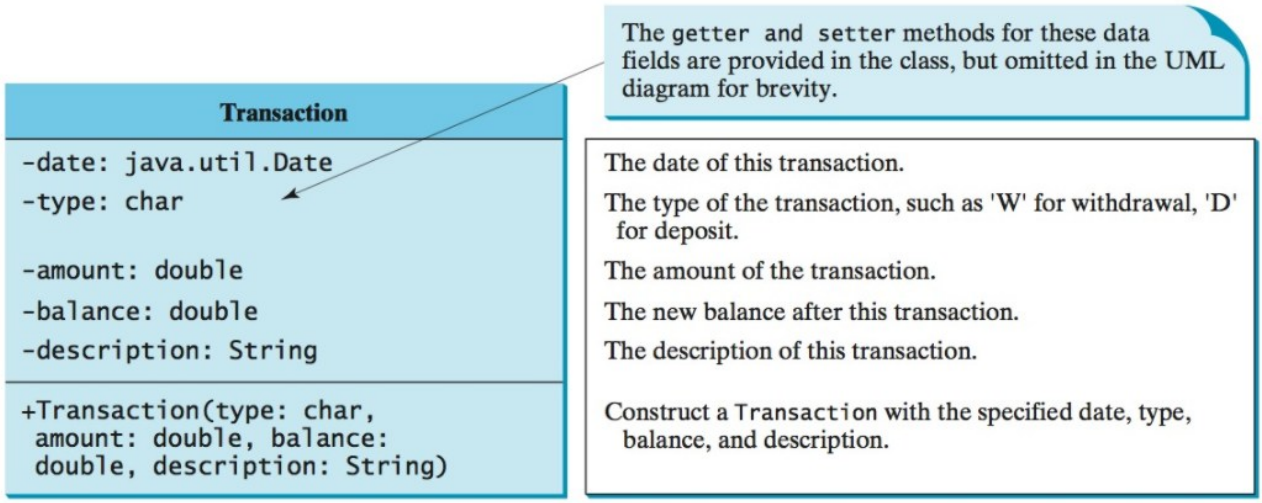
1. The Account class was defined to model a bank account. An account has the properties account number, balance, annual interest rate, and date created, and methods to deposit and withdraw funds.

    a. Create two subclasses for checking and saving accounts. A checking account has an overdraft limit with default value 5000, but a savings account cannot be overdrawn.

    b. Write suitable withdraw() methods for checking and savings account.

    c. Draw the UML diagram for the classes and then implement them. Write a test program that creates objects of SavingsAccount, and CheckingAccount and invokes their toString() methods.

2. (New Account class)
   Design a new Account class as follows:
   - Add a new data field name of the String type to store the name of the customer.
   - Add a new constructor that constructs an account with the specified name, id, and balance.
   - Add a new data field named transactions whose type is ArrayList that stores the transaction for the accounts. Each transaction is an instance of the Transaction class. The Transaction class is defined as shown above:



   - Modify the withdraw and deposit methods to add a transaction to the transactions array list.
   - All other properties and methods are the same as in Programming Exercise 9.7.
   - Write a test program that creates an Account with annual interest rate 1.65%, balance 1000, id 1122, and name George. Deposit $30, $40, and $50 to the

account and withdraw $5, $4, and $2 from the account. Print an account summary that shows account holder name, interest rate, balance, and all transactions.

3. (Smallest rows and columns) Write a program that randomly fills in 0s and 1s into an n-by-n matrix, prints the matrix, and finds the rows and columns with the most 0s. (Hint: Use two ArrayLists to store the row and column indices with the most 0s.)

```
Enter the array size n: 4 Enter
The random array is
1110
0111
0110
0100
The most 0s row index: 3
The most 0s column index: 0, 3
```